Asset Growth and the Cross-Section of Stock Returns

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Introduction

Main Goal

- Main Goal: Test for **firm-level asset investment effects in returns** by examining the crosssectional relation between **firm asset growth** and **subsequent stock returns**.

Inspiration

- A bias in the market's capitalization of corporate asset investment and disinvestment.
- Corporate events associated with **asset expansion** (i.e., acquisitions, public equity offerings, public debt offerings) tend to have **low returns**
- Corporate events associated with **asset contraction** (i.e.share repurchases, debt prepayments, and dividend initiations) tend to have **high returns**.
- Other events (i.e. capital investment, accruals, sales growth rates) are found to be **negatively correlated with future returns**

This paper uses a single, comprehensive measure that captures both asset expansion and contraction: a firm's total asset growth rate.

Paper Structure

- Section I: Data and Characteristics of High- and Low-Growth Firms
- Section II: The Effect of Asset Growth on Future Returns
- Section III: Is the Asset Growth Effect Due to Risk or Mispricing?

Section I: Data and Characteristics of High- and Low-Growth Firms

Data and Key Variable

Sample

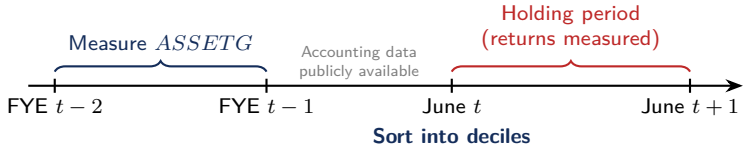
- All NYSE, Amex, and NASDAQ firms
- CRSP (monthly stock returns) + Compustat (annual accounting data)
- Excludes financial firms (SIC 6000–6999)
- Return period: July 1968 – June 2003
- Firm must be on Compustat for 2 years

Main variable: annual asset growth

$$ASSETG_t = \frac{TA_{t-1} - TA_{t-2}}{TA_{t-2}}$$

- TA = total assets (Compustat item 6)
- Year-on-year % change in total assets
- Captures *all* forms of asset expansion and contraction in one simple measure

Portfolio Formation and Timing



- End of June t: sort stocks into **10 deciles** by $ASSETG$ (Decile 10 $\Rightarrow$ highest, Decile 1 $\Rightarrow$ lowest $ASSETG$)
- Hold July t – June $t+1$; rebalance annually and repeat over 35 years (1968–2003)
- **Size groups** (if needed): small / medium / large using NYSE 30th and 70th percentiles; Deciles formed *within* each size group ($3 \times 10 = 30$ portfolios in total)
- High growth rate firms $\neq$ Large Capitalization firms

Equal-weighted (EW)

$$R_{p,t}^{EW} = \frac{1}{N_p} \sum_{i=1}^{N_p} R_{i,t}$$

Value-weighted (VW)

$$R_{p,t}^{VW} = \sum_{i=1}^{N_p} w_{i,t-1} \, R_{i,t}$$

$$w_{i,t-1} = \frac{MV_{i,t-1}}{\sum_{j=1}^{N_p} MV_{j,t-1}}$$

Characteristics of High- vs. Low-Growth Firms

Table I

Asset Growth Deciles: Financial and Return Characteristics

At the end of June of each year t over 1968 to 2002, stocks are allocated into deciles based on asset growth (*ASSETG*) defined as the percentage change in total assets from the fiscal year ending in calendar year $t-2$ to fiscal year ending in calendar year $t-1$. The table reports the financial and return characteristics in the year prior to the portfolio formation date. *L2ASSETG* is the asset growth defined as the percentage change in total assets from the fiscal year ending in calendar year $t-3$ to fiscal year ending in calendar year $t-2$. *ASSETS* is Compustat data item 6, total assets, in millions of $, from the fiscal year ending in calendar year $t-1$. Market value (*MV*), in millions of $, is calculated using the price and the number of shares outstanding at the end of June of year t. All accounting variables (book-to-market ratio (*BM*), earnings-to-price ratio (*EP*), leverage, return on assets (*ROA*), and *ACCRUALS*) are calculated using Compustat data in the fiscal year ending in calendar year $t-1$. *BHRET6* is the buy-and-hold return over January(t) to June(t) where t is the portfolio formation year. *BHRET36* is the 36-month buy and hold return over July($t-3$) to June(t). *ISSUANCE* is a 5-year change in the number of equity shares outstanding. The numbers in each cell are time-series averages of yearly cross-sectional medians, with the exception of average market value (*MV-AVG*), in millions of $, which is the time-series average of yearly cross-sectional mean capitalization. All numbers, with the exception of *ASSETS*, *MV*, and *MV-AVG*, are in decimal form, that is 0.01 is 1%. Details on the construction of these variables are provided in the Appendix.

Decile	ASSETG	L2ASSETG	ASSETS	MV	MV-AVG	BM	EP	Leverage	ROA	BHRET6	BHRET36	ACCRUALS	ISSUANCE
1 (Low)	−0.2115	0.0041	20.86	15.70	130.27	0.8156	−0.1931	0.2446	−0.0186	0.086	−0.3286	−0.1253	0.0803
2	−0.0679	0.0294	48.35	26.64	276.70	1.0266	−0.005	0.2437	0.0704	0.0917	−0.1121	−0.0700	−0.0353
3	−0.0079	0.0447	96.46	53.98	602.21	0.9974	0.0536	0.2485	0.1110	0.0921	0.0647	−0.0487	−0.0841
4	0.0319	0.0601	134.48	84.92	930.28	0.9261	0.0732	0.2429	0.1329	0.0784	0.1964	−0.0356	−0.1139
5	0.0661	0.0748	154.60	106.88	1154.12	0.8515	0.0797	0.2400	0.1473	0.0792	0.2757	−0.0283	−0.104
6	0.1025	0.0950	145.20	112.72	1123.79	0.7710	0.0808	0.2263	0.1608	0.0703	0.3360	−0.0206	−0.0752
7	0.1480	0.1165	130.18	116.69	1051.26	0.6918	0.0794	0.2139	0.1743	0.0662	0.3804	−0.0113	−0.0295
8	0.2168	0.1465	100.71	103.96	889.45	0.6076	0.0759	0.2181	0.1886	0.0622	0.4713	0.0014	0.0144
9	0.3529	0.1839	76.74	95.65	903.01	0.5158	0.0685	0.2356	0.2025	0.0427	0.5657	0.0119	0.1167
10 (High)	0.8357	0.2168	66.69	85.61	585.35	0.4256	0.0528	0.2612	0.2173	0.0074	0.7154	0.0341	0.3012
Spread (10-1)	1.0471	0.2127	45.83	69.91	455.08	−0.39	0.246	0.0165	0.2359	−0.0786	1.044	0.1594	0.2209
t(spread)	15.60	26.26	4.83	5.24	3.32	−6.48	5.61	1.17	20.64	−0.33	15.65	28.10	8.36

Characteristics of High- vs. Low-Growth Firms

Characteristic	High (D10)	Low (D1)	Interpretation
Financial characteristics			
Book-to-market (BM)	0.43	0.82	Growth stocks vs. value stocks
Earnings-to-price (EP)	0.05	-0.19	Low-growth firms unprofitable on average
ROA	21.7%	-1.9%	High-growth firms more profitable
Accruals	3.4%	-12.5%	Earnings exceed cash flow (lower earnings quality)
Share issuance (5-yr)	30.1%	8.0%	Issue more equity to fund growth
Leverage	0.26	0.24	Similar leverage
Past stock performance			
Past 36-month return	71.5%	-32.9%	Strong run-up over prior 3 years
Past 6-month return	0.7%	8.6%	Below average recently

Source: Table I. All D10−D1 spreads are significant except leverage and past 6-month return. Corr($ASSETG$, BM) $= -0.18$.

Takeaway

At formation, high-growth firms look like **winners**: strong past returns, high profitability, low BM, and heavy equity issuance.

$\Rightarrow$ Past success likely leads firms to pursue aggressive growth (Jensen, 1986).

Section II: The Effect of Asset Growth on Future Returns

Does $ASSETG$ predict stock returns?

$ASSETG$ and BM:

- $ASSETG$ side: High-growth firms have low BM; Value effect: High BM has higher returns and low BM has lower returns
- How to identify the predictability comes from $ASSETG$ or BM when they have the same direction?

Fama–French three-factor alpha: return *not* explained by known risk factors

$$R_{p,t} - R_{f,t} = \alpha_p + \beta_p \, MKT_t + s_p \, SMB_t + h_p \, HML_t + \varepsilon_{p,t}$$

- Null: the three-factor model adequately prices growth-related risk $\Rightarrow \alpha_p = 0$
- Significant $\alpha_p \Rightarrow$ **preliminary evidence of mispricing**

Main Result: Returns in the Year After Sorting

		D1 (Low)	D10 (High)	10−1	t-stat
Raw return	EW	1.99%	0.26%	−1.73%	−8.45
	VW	1.48%	0.43%	−1.05%	−5.04
FF3 alpha	EW	0.76%	−0.87%	−1.63%	−8.33
	VW	0.24%	−0.46%	−0.70%	−3.84

Monthly returns, July 1968 – June 2003. Source: Table II, Panels B and C (Year 1, all firms).

- Returns **decline monotonically** across all 10 deciles (EW: 1.99% → 1.76% → ⋯ → 0.85% → 0.26%)
- Negative spread ⇒ high-growth firms underperform low-growth firms
- Low-growth firms earn **positive** alphas; high-growth firms earn **negative** alphas
- Significant α spread even after controlling for $HML \Rightarrow ASSETG$ contains information beyond the value effect
- Risk-adjusted spread ≈ **20% per year (EW)** and **8% per year (VW)**

Robust Across Firm Size and Subperiods

FF3 alpha spread (10−1)	**EW**	**VW**
By size group		
All firms	−1.63% (−8.33)	−0.70% (−3.84)
Small	−1.77% (−9.12)	−1.14% (−6.46)
Medium	−0.60% (−2.85)	−0.55% (−2.45)
Large	−0.86% (−3.12)	−0.81% (−2.91)
By subperiod		
1968–1980	−0.71% (−3.87)	−0.35% (−1.69)
1981–1990	−1.34% (−4.80)	−0.66% (−2.51)
1991–2003	−2.70% (−6.62)	−1.07% (−2.52)

Monthly alphas in Year 1; t-statistics in parentheses. Source: Table II, Panels C and D.

- Significant in **all size groups**: large firms $\approx$ 0.81% per month (10% per year)
  $\Rightarrow$ not driven by small, illiquid stocks
- Significant in **every subperiod** and **strongest in 1991–2003**
  $\Rightarrow$ the effect is not disappearing over time

Before vs. After Sorting: A Sharp Reversal

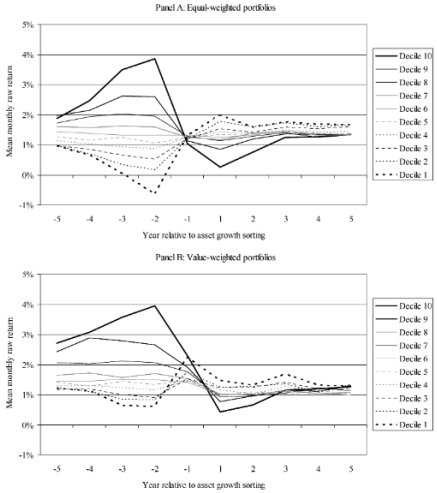


Figure 2. Mean monthly raw returns in event time

Bold solid: Decile 10 (High) Bold dashed: Decile 1 (Low)

Before sorting (years -3, -2)

- High-growth firms earn **much higher** returns (year -2 spread: EW $+4.50\%$, VW $+3.34\%$ per month)

$\Rightarrow$ Past success precedes expansion

After sorting (year 1)

- Pattern **reverses**: high-growth firms earn the **lowest** returns
- Spread (10$-$1): EW -1.73% ($t = -8.45$), VW -1.05% ($t = -5.04$)

Persists for years

- Cumulative 5-year spread: EW -88% ($t = -8.63$), VW -50% ($t = -4.25$)
- EW significant in all 5 years; VW through year 3

Source: Table II, Panel B; Figure 2.

Cross-Sectional Regression: Partial out other effects

Concern: high-growth firms also have low BM, strong past returns, high accruals, and high investment (Table I)

$\Rightarrow$ Is the asset growth effect just a **known anomaly in disguise**?

Method: Fama–MacBeth (1973) regressions

1. Each year, run a cross-sectional regression of annual returns (July t – June $t+1$):
$$R_i = \gamma_0 + \gamma_1 \, ASSETG_i + \textstyle\sum_k \gamma_k \, X_{k,i} + \varepsilon_i$$
2. Average each γ over 35 years; t-statistics adjusted for autocorrelation(avoid overstating significance)

$\Rightarrow$ Tests the effect of $ASSETG$ **holding other variables constant**

Base controls

- BM, size (Fama–French, 1992)
- 6-month return: momentum
- 36-month return: long-term reversal

Other growth measures

- 5-year sales growth (LSV, 1994)
- Capital investment (TWX, 2004)
- NOA/A (Hirshleifer et al., 2004)
- Accruals (Sloan, 1996)

Asset Growth vs. Standard Predictors

t-statistics	All firms	Small	Medium	Large
$ASSETG$	$\mathbf{-6.52}$	$\mathbf{-5.18}$	$\mathbf{-3.80}$	$\mathbf{-3.60}$
BM	3.40	2.20	1.50	0.58
Size (MV)	-1.57	-1.47	-1.26	-0.84
6-month return	1.09	0.72	1.63	1.55
36-month return	0.57	-0.20	1.42	2.21

Fama–MacBeth regressions of annual returns, 1968–2003 (Model 1). Source: Table III.

- $ASSETG$ is the **strongest predictor**: t-stat about **twice** that of BM
- Negative coefficient (-0.092): higher asset growth $\Rightarrow$ lower future returns
- Significant in **every size group**; BM and size lose significance among large firms

Asset Growth vs. Other Growth Measures

	All firms		Large firms	
Added variable	$ASSETG$	Added	$ASSETG$	Added
5-year sales growth	-7.41	-0.27	-4.59	-0.68
Capital investment	-6.05	-3.32	-3.23	-0.41
NOA/A	-6.10	-2.43	-3.76	-2.54
Accruals	-5.24	-4.00	-2.39	-0.96
5-year asset growth	-6.98	-2.22	-3.79	-1.07

t-statistics; each row adds one growth variable to the base model. Source: Table III, Panels A and D (Models 3–7).

- $ASSETG$ remains significant in **every model and size group**
- Among large firms, sales growth, capital investment, accruals, and 5-year growth are insignificant $\Rightarrow$ $ASSETG$ **subsumes** most other growth measures

Takeaway

Asset growth is not a proxy for known predictors; it is the **single strongest predictor** of the cross-section of returns.

Decomposing Asset Growth: The Balance Sheet View

Question: which parts of asset growth drive the return effect?
Is it only **equity-financed** growth (i.e., managers timing the market)?

Investment side (assets)	**Financing side (liabilities & equity)**
ΔCash	ΔOperating liabilities
ΔNon-cash current assets	ΔRetained earnings
ΔPP&E (Fixed assets)	ΔStock financing
ΔOther assets	ΔDebt financing
$= ASSETG$	$= ASSETG$

Each change is scaled by lagged total assets, so the components sum exactly to $ASSETG$.

Decomposition Regressions: Method

Variables

$$\Delta X_i = \frac{X_{i,t-1} - X_{i,t-2}}{TA_{i,t-2}}$$

$X \in \{Cash,\ CurAsst,\ PPE,\ OthAssets,\ OpLiab,\ RE,\ Stock,\ Debt\}$; TA: total assets
R_i: annual return of firm i from July t to June $t+1$

(1) Alone: one component at a time ⇒ does it predict returns by itself?

$$R_i = \gamma_0 + \gamma_1\, \Delta Component_i + \varepsilon_i$$

(2) Joint: all four components of one side ⇒ which still matters after controlling for the others?
Investment side:

$$R_i = \gamma_0 + \gamma_1 \Delta Cash_i + \gamma_2 \Delta CurAsst_i + \gamma_3 \Delta PPE_i + \gamma_4 \Delta OthAssets_i + \varepsilon_i$$

Financing side:

$$R_i = \gamma_0 + \gamma_1 \Delta OpLiab_i + \gamma_2 \Delta RE_i + \gamma_3 \Delta Stock_i + \gamma_4 \Delta Debt_i + \varepsilon_i$$

- **Fama–MacBeth**: estimate each year, average γ's over 35 years; t-stats adjusted for autocorrelation
- $\gamma_k < 0$ and significant ⇒ growth in component k predicts **lower** returns
- Two sides estimated **separately** (each sums to $ASSETG$); repeated by firm size

Which Components Matter? (All Firms)

Investment side			Financing side		
	Alone	Joint		Alone	Joint
ΔCash	-0.03	0.19	ΔOp. liab.	-4.00	-0.99
ΔCur. assets	-4.80	-3.74	ΔRet. earn.	-0.83	-0.91
ΔPP&E	-3.91	-2.76	ΔStock	-1.88	-2.13
ΔOther	-3.34	-1.95	ΔDebt	-6.59	-5.01

t-statistics; red = significant at 5%. Alone: one component per regression. Joint: all four components of that side together.

Source: Table IV, Panel A.

- **Investment side**: operating assets (current assets, PP&E) matter most; **cash does not**
- **Financing side**: external financing (debt, stock) matters most; **retained earnings do not**
- Debt has the strongest effect $\Rightarrow$ **not just equity market timing**
- When $ASSETG$ and any single component enter together, $ASSETG$ is the strongest

The Key Component Shifts with Firm Size

Joint regression t-stats	Small	Medium	Large
Investment side			
ΔNon-cash current assets	**-4.93**	-2.39	-0.22
ΔPP&E	-2.11	-1.46	**-2.64**
Financing side			
ΔDebt	**-4.09**	-3.90	-1.15
ΔStock	-2.33	-1.45	**-2.61**

Bold red = key component in that size group. Cash and retained earnings are insignificant in all size groups. Source: Table IV, Panels B–D.

- **Small firms**: current assets and debt financing
- **Large firms**: PP&E and stock financing

Why does total asset growth work so well?

$ASSETG$ **aggregates all components**, so it captures the return predictability of each one, even as their relative importance shifts across firm sizes.

Section III: Is the Asset Growth Effect Due to Risk or Mispricing?

Can Risk Explain the Asset Growth Effect?

Risk-based theory (Berk, Green & Naik, 1999; Zhang, 2005; and others)

- Investing converts risky **growth options** into safer **assets in place**

$\Rightarrow$ High-growth firms(=more investing) $\Rightarrow$ less risky $\Rightarrow$ investors expected **lower expected returns**

But the evidence is inconsistent with this story

1. **Too large**: FF3 alpha spread of 8% (VW) to 20% (EW) per year, although the theory implies size and BM should capture this risk
2. **Zero risk premium**: high-growth firms earn only 5.2% (VW) and 3.1% (EW) per year, below the 6.3% risk-free rate
3. **Wrong pattern over time**: although high-growth firms continue to grow in years 1–5, their returns **increase** over the same period

Evidence for Mispricing

1. Earnings announcement returns

- If risk: returns similar on announcement and other days
- If mispricing: investors are **surprised** when earnings are released

Daily return, year 1	**EA days**	**Other days**	**Difference (t)**
Low growth (D1)	0.27%	0.11%	$+0.16\%$ (3.41)
High growth (D10)	-0.10%	0.00%	-0.10% (-3.59)

$\Rightarrow$ High-growth firms get **bad surprises**; low-growth firms get **good surprises**

2. Profitability reverses

- Operating margin (EBITDA/Sales) around the sorting year (Figure 4):
- Firms grow after unusually good performance and shrink after unusually bad performance, but performance later mean-reverts

$\Rightarrow$ Investors **over-extrapolate** past performance: too optimistic about high-growth firms, too pessimistic about low-growth firms

Figure 4

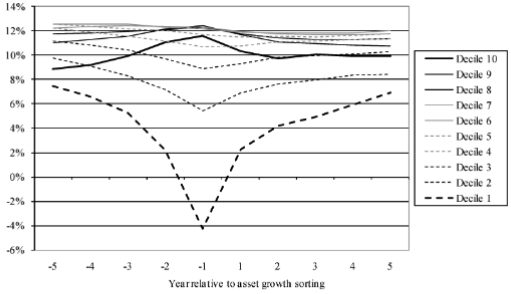


Figure 4. Mean equal-weighted operating margin for asset growth deciles in event time. At the end of June of each year from 1968 to 2002, stocks are allocated into deciles based on asset growth rates defined as the percentage change in total assets from the fiscal year ending in calendar year $t-2$ to fiscal year ending in calendar year $t-1$. The time-series mean of the annual median decile operating margin statistics is reported every year for 10 years around the portfolio formation year (t) over the period of July 1968 to June of 2003. Operating margin is defined as operating income before depreciation (Compustat data item 13) divided by contemporaneous net sales (Compustat data item 12). Mean operating margin for the portfolios in the year following portfolio formation is reported on the x-axis labeled as Year 1, over the next year is reported on the x-axis labeled as Year 2, and over the contemporaneous sorting period is reported on the x-axis labeled as Year -1.

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Thank You